Priyanshu Doshi

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Education

Institute of Technology, Nirma University *Jul 2024 – Jul 2028*
B.Tech in Artificial Intelligence and Machine Learning
CGPA: 8.85/10
Class XII (Science): 96.7% — 99.1 Percentile — Rank 160 (Gujarat) *2024*

Skills Summary

Programming: Python, C++, C, JavaScript, HTML, CSS
Machine Learning & AI: PyTorch, TensorFlow, Scikit-Learn, XGBoost, SHAP, NumPy, Pandas, Matplotlib, Optuna, LLM APIs, LangChain, RAG, Prompt Engineering, FAISS, SentenceTransformers, LangSmith
Web & Backend: FastAPI, Express.js, Next.js, React, TailwindCSS, REST APIs
Databases & Cloud: MySQL, AWS RDS, Google Cloud Platform (GCP)

Experience

Full Stack Developer Intern - MZHUB Faithtech (Remote) *Dec 2025 - Dec 2025*
- Developed and deployed a production-grade Next.js web platform on Microsoft Azure App Service with responsive and scalable frontend architecture.

AI/ML Intern — Elevate Labs (Remote) *May 2025 – June 2025*
- Top-performing AI/ML intern; built end-to-end NLP and Computer Vision pipelines using PyTorch, TensorFlow, and Scikit-learn for model training and evaluation.

Vice Chair — ACM Student Chapter, Nirma University *Sep 2025 – Present*
- Led technical workshops, hackathons, and industry events for students.

Research Experience

Robotic Arm Fault Detection using CatBoost Classifier *(IEEE Sensors Letters, 2026)*
- Proposed a CatBoost-based classification model achieving 97.20% accuracy and a 0.9718 F1-score on the CASPER dataset.
- Outperformed traditional machine learning methods including SVM, Logistic Regression, Naive Bayes, and QDA for industrial robotic fault detection.

Projects

Spectra Scan — Automated Paint Defect Detection System
- Contributed to development of a 2m × 2m × 2m gantry-based automated system for automobile paint defect inspection.
- Developed a centralized control application for operating and monitoring the complete inspection system.
- Implemented machine vision analytics for defect detection, along with a RAG-based troubleshooting assistant and WebAR visualization for industrial system monitoring and diagnostics.

Lumin.AI — AI-Powered Solar Plant Risk Monitoring Platform (AUBERGINE track)
- Built a full-stack solar inverter monitoring and failure prediction platform using XGBoost with SHAP explainability, integrated with a Next.js dashboard, Express.js APIs, and AWS RDS.
- Developed a GenAI microservice that converts ML predictions and SHAP features into operator guidance using Groq Llama 3.3 70B, implementing RAG with SentenceTransformers + FAISS and structured prompt outputs.
- Designed agentic workflows for automated maintenance ticket generation (JSON + PDF), implemented 4-layer hallucination guardrails with LangSmith tracing, and conducted LLM ablation across Groq Llama 3.3 70B, Gemini 2.5 Flash, and Qwen 2.5 72B.

SENTINEL — Privacy-Preserving Edge AI Mental Wellbeing System
- Architected an Edge AI wellbeing system on Raspberry Pi 5 using Phi-3 Mini, Ollama, FER, VADER, and ChromaDB for fully offline multimodal inference.
- Engineered a multimodal Edge ML pipeline integrating facial emotion recognition, sentiment fusion, and RAG-based contextual memory retrieval.
- Developed a low-latency Flask + SSE streaming architecture for real-time on-device LLM inference and emotion-aware interaction.

Achievements

- **Winner – Aubergine Track, HackAMined National Hackathon** (2026) – Secured 1st place in the Aubergine track and ranked **Top 5 Overall** (Finalist) among **400+ teams and 2300+ participants** in a national-level hackathon.
- **Winner – CodeAdda Premier League (Codeforces)** (Apr 2025)
- **Reliance Foundation Scholar** (2025) – Awarded Reliance Foundation Undergraduate Scholarship for academic excellence.
- **National Rank 4 – Mitsubishi Electric Cup (6th Edition)** (2026) – Secured **4th rank nationally** in a national-level competition focused on Factory Automation.

Certifications

- **Supervised Machine Learning** - DeepLearning.AI (Stanford) - Coursera — Score: 99.83% — Certificate
- **PyTorch for Deep Learning Bootcamp** - Udemy — Certificate